Student Print Packet - REVISED

PV101 Day 5 - Monday, July 13, 2026

Name: _____ Date: _____

Morning Quiz

1. In a solar/battery circuit, what does voltage describe?

Answer: _____

2. What does current describe?

Answer: _____

3. What is the formula for power in watts?

Answer: _____

4. A 12.8 V battery is supplying 5 A. About how many watts is that?

Answer: _____

5. A 600 W inverter load on a 12 V battery can require roughly how much current before losses?

Answer: _____

6. In series, what usually adds: voltage or current capacity?

Answer: _____

7. In parallel, what usually stays the same and what can increase?

Answer: _____

8. What is a fuse actually protecting?

Answer: _____

9. Where should the main load/inverter fuse be placed?

Answer: _____

10. Why measure battery voltage before connecting equipment?

Answer: _____

11. What is voltage drop?

Answer: _____

12. Name one sign that a connection should be treated as unsafe.

Answer: _____

Afternoon Lab - Commissioning & Fault Finding

Team/Kit: _____ Students: _____

Commissioning record

Battery open-circuit voltage: _____ V
Controller battery voltage: _____ V
PV/input voltage: _____ V
Load voltage at source end: _____ V
Load voltage at load end: _____ V
Voltage drop: _____ V
Instructor initials: _____

Fault record

Observed symptom: _____
Prediction: _____
Measurement evidence: _____
Fault identified: _____
Corrective action: _____
Proof of restoration: _____

Wire / fuse reasoning

Load power: _____ W
System voltage: _____ V
Estimated current: _____ A
Proposed conductor: _____
Proposed fuse: _____
Why this protects the wire: _____

Serviceability handoff

Label added or corrected: _____
Useful future test point: _____
Note for next technician: _____

Exit ticket

One reading that proved something: _____
One fault I can diagnose systematically: _____
One reason we did not rebuild the system: _____